

AWS CloudTrail Incident Investigation Report

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Project: CloudTrail Incident Investigation Lab

Executive Summary

This investigation analyzes suspicious activity generated by the user account 'Attacker-user' within an AWS environment. CloudTrail logs indicate successful console access, reconnaissance activity, resource discovery, CloudTrail inspection, and attempts to modify logging-related infrastructure. The objective was to identify attacker actions, build a timeline, and assess impact.

Scope

Evidence reviewed included CloudTrail Event History screenshots, IAM activity, S3 access attempts, CloudTrail event details, and the supplied timeline.

Key Findings

- Multiple successful console logins were observed.
- Resource reconnaissance was performed against IAM, EC2, S3 and Resource Explorer.
- CloudTrail configuration was investigated by the attacker.
- Attempts to access or modify resources generated Access Denied events.
- A trail-related modification activity was observed and subsequently verified by the actor.

Incident Timeline (Source Material)

Incident Timeline

Initial Access & Persistent Sign-In Attempts

11:05:03 — 11:25:26

The user account Attacker-user logs into the AWS Management Console (ConsoleLogin). There are three logged logins over a 20-minute window (11:05, 11:11, and 11:25), indicating potential session refreshes or multiple concurrent logons.

Initial Reconnaissance Sweep

11:25:36 — 11:25:38

Immediately after the final successful login, the attacker performs a rapid, automated discovery sweep across AppRegistry (ListApplications), EC2 regional configurations (DescribeRegions), and Notification Hubs (ListNotificationHubs). They are mapping out where infrastructure is currently deployed.

Deep Discovery via AWS Resource Explorer

11:26:50 — 11:28:48

The attacker pivots heavily to AWS Resource Explorer 2. Instead of querying services one by one, they run repeated ListIndexes and Search queries. This allows them to discover all active resources across multiple AWS regions simultaneously. They wrap this up by listing S3 buckets (ListBuckets) at 11:28:11.

Targeting Defensive Controls & Infrastructure Modification

11:30:03 — 11:31:02

The attacker begins a massive discovery spike targeting AWS CloudTrail, checking existing trails (ListTrails, DescribeTrails) and digging through log events (LookupEvents).

At 11:31:02, the attacker makes an impactful infrastructure move: they execute CreateBucket to create an S3 bucket named aws-cloudtrail-logs-238480774707-0d220714. This naming convention is often used by attackers attempting to blend in or redirect logging infrastructure to cover their tracks.

Post-Modification Log Verification

11:31:53

The attacker immediately loops back to CloudTrail, running multiple DescribeTrails, LookupEvents, and ListEventDataStores API calls. They are highly likely checking to see if their bucket creation altered the logging paths or auditing infrastructure.

Final Reconnaissance Round

11:32:07 — 11:32:15

The timeline closes out with a rapid final sweep: querying Resource Explorer (Search, ListIndexes), checking EC2 configurations (DescribeRegions), and performing a final ListBuckets check on S3 at 11:32:15.

SOC Analyst Assessment

The behavior aligns with a typical cloud intrusion lifecycle: initial access, environment discovery, defensive control enumeration, and validation of changes. CloudTrail provided sufficient visibility to reconstruct the sequence of events.

Recommendations

1. Enable multi-region CloudTrail logging.
2. Configure log integrity validation and centralized storage.
3. Apply least-privilege IAM permissions.
4. Monitor CloudTrail modification events with alerts.
5. Investigate all AccessDenied events for privilege escalation attempts.

Project Rating

Portfolio Level: Beginner to Intermediate Cloud SOC Project. The project demonstrates AWS logging analysis, timeline reconstruction, and incident response fundamentals.